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Hardware

Nano RP2040 Connect



Tutorials

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This tutorial refers to a product that has reached its end-of-life status.

Nano RP2040 Datalogger with MicroPython

Learn how to save data in .csv format on the Nano RP2040 Connect, using MicroPython.

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Introduction

The [Nano RP2040 Connect](#) board has on-board storage that allows you to turn the device into a data logger without any extra components.

In order to utilize this feature, we need to install the latest release of [OpenMV's flavor of MicroPython](#).

Goals

The goals of this tutorial are:

Install OpenMV MicroPython firmware on the board.

Learn how to save data in a `.csv` format directly on the Nano RP2040 Connect.

[Help](#)

Hardware & Software Needed

[Thonny IDE.](#)

[Arduino Nano RP2040](#)

[Connect.](#)

Install MicroPython

1. The first step is to install the latest version of the OpenMV firmware (MicroPython) on the Nano RP2040 board. This version is available through the link below:

[OpenMV firmware v4.3.1. \(download\)](#)

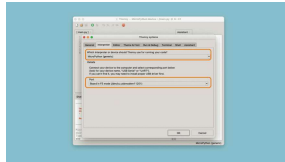
2. Unzip the contents, and locate the `firmware.uf2` file inside of the **ARDUINO_NANO_RP2040_CONNECT** folder.

3. Force the bootloader on the Nano RP2040 Connect, by connecting a jumper wire between `GND` and `REC` pins as shown in the image below. When the mass storage device opens, drag the `firmware.uf2` onto it, and the latest version will install.



Connect GND + REC +
press the reset button
to open mass storage.

4. Install and open the **Thonny IDE**. Navigate to **Run > Select Interpreter** and choose the **"MicroPython(generic)"** from the list. Your board should now appear in the other dropdown menu:



Thonny board/port selection.

If your board appears, it has been successful. In this case, it is called

```
Board in FS mode  
(/dev/cu.usbmodem11201)
```

Data Logger Example

Now that the OpenMV MicroPython firmware is installed on your device, and it is detected using Thonny, we can create our datalogger.

The script for the datalogger is quite basic, and has the following functionality:

Create a `.csv` file

Read the value of an analog pin, and log it, using the `file.write()` function.

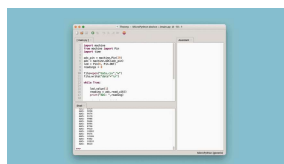
Repeat 25 times and then finish script.

Each time a reading is recorded, the built-in LED flashes.

The script for this tutorial can be found below:

```
1 import machine
2 from machine import Pin
3 import time
4
5 adc_pin = machine.Pin(2, machine.Pin.IN)
6 adc = machine.ADC(adc_pin)
7 led = Pin(6, Pin.OUT)
8 readings = 0
9
10 # create a file named data.csv
11 file=open("data.csv", "a")
12 file.write("data"+str(readings)+"\n")
13
14 while True:
15     led.value(1)
16     reading = adc.read_u16()
17     print("ADC: ", reading)
18     time.sleep_ms(100)
19
20     # convert and write to file
21     file.write(str(reading)+"\n")
22
23     led.value(0)
24     time.sleep_ms(100)
25     readings += 1
26
27 # Close the file
28 file.close()
```

Copy paste this code into the Thonny editor, and click on the **Green Play Button (F5)**. The values recorded are also printed in the terminal, so we can compare it later. After running, it should look like this:



Thonny IDE, data
printed in terminal

When you run the script, the board should now start blinking fast, every 100 milliseconds, and it will do so 25 times (as is specified in the code, the number can be changed).

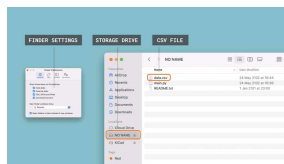
Accessing Data

Once done, navigate to Finder / Explorer, and locate a drive called **"NO NAME"**. This should now include a `data.csv` file. This contains the 25 readings we just made by running the script.

If you are using a Mac, you may need to change a setting that allows you to see external disks. If you can't see the drive, go to Finder > Preferences and tick the boxes that appear.



see external disks. If you can't see the drive, go to Finder > Preferences and tick the boxes that appear.



The "NO NAME" drive with a data.csv file.

Congratulations, you have now successfully recorded data and stored it in a `.csv` file onboard the Nano RP2040 Connect.

Please note that you should never open this file whenever a file management operation is ongoing. This will most likely corrupt your file and you won't be able to obtain the data. Best practice is to record the data, wait a little, and then open up the `data.csv` file.

Conclusion

In this tutorial, we turned a Nano RP2040 Connect board into a data logger, without the use of any external components (such as an SD card).

This is an incredibly useful tool whenever you are working with data collection, and the script found in this tutorial can be easily be altered to fit your project.

While we in this tutorial only recorded values from an analog pin, there are many other things to do, such as:

- Record data from the onboard IMU.

- Record data from an external sensor.

- Record events that occurred along with a timestamp (for example, how many times a sensor value's threshold was met).

no.cc is
facilitated
through a
public
GitHub
repository.
If you see
anything
wrong, you
can edit
this page
here.

Discover
Arduino
Discord

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